

WARP Violation: Different Danger Zone Scenarios

Complete Visual Guide

February 28, 2026

1 Overview

The WARP violation "danger zone" is defined by TWO inequalities:

1. **Old budget constraint:** $p_1^{old}x_1 + p_2^{old}x_2 \leq m$ (was affordable before)
2. **New iso-cost constraint:** $p_1^{new}x_1 + p_2^{new}x_2 \geq C$ (costs enough now)

The shape and existence of the danger zone depends on:

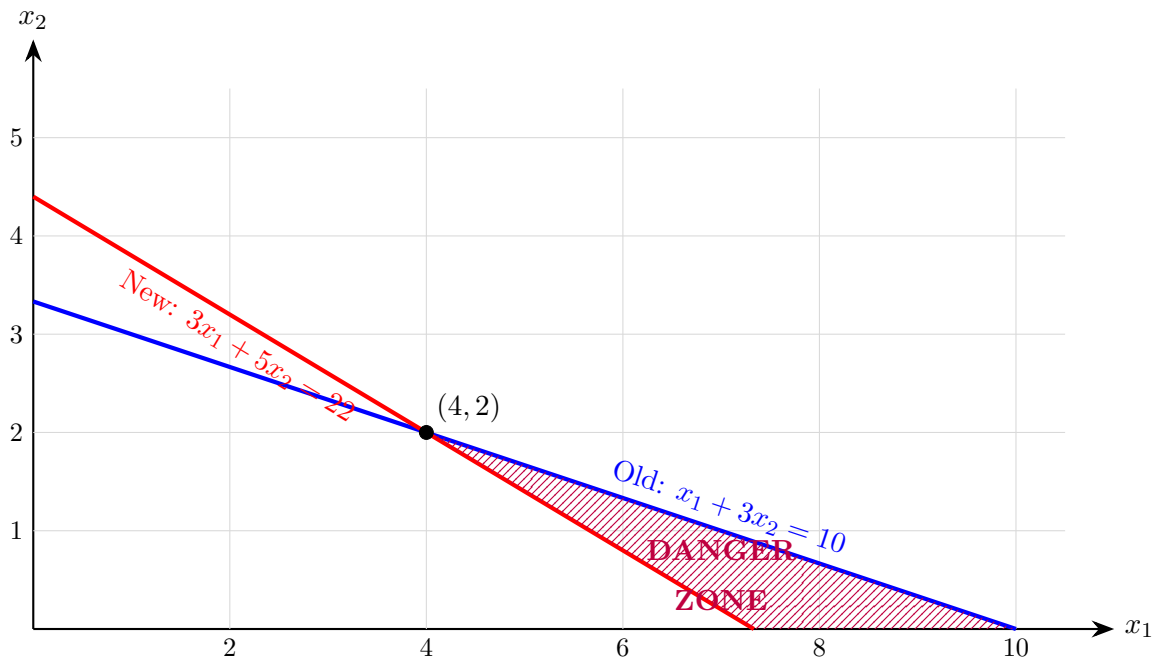
- The slope of each line
- Where they intersect (if at all)
- The relative price changes

Let's explore different scenarios!

2 Scenario 1: Triangle Touching x-axis (Classic Case)

Setup:

- Old prices: $(1, 3)$, bought $(4, 2)$, budget = 10
- New prices: $(3, 5)$, old bundle costs 22



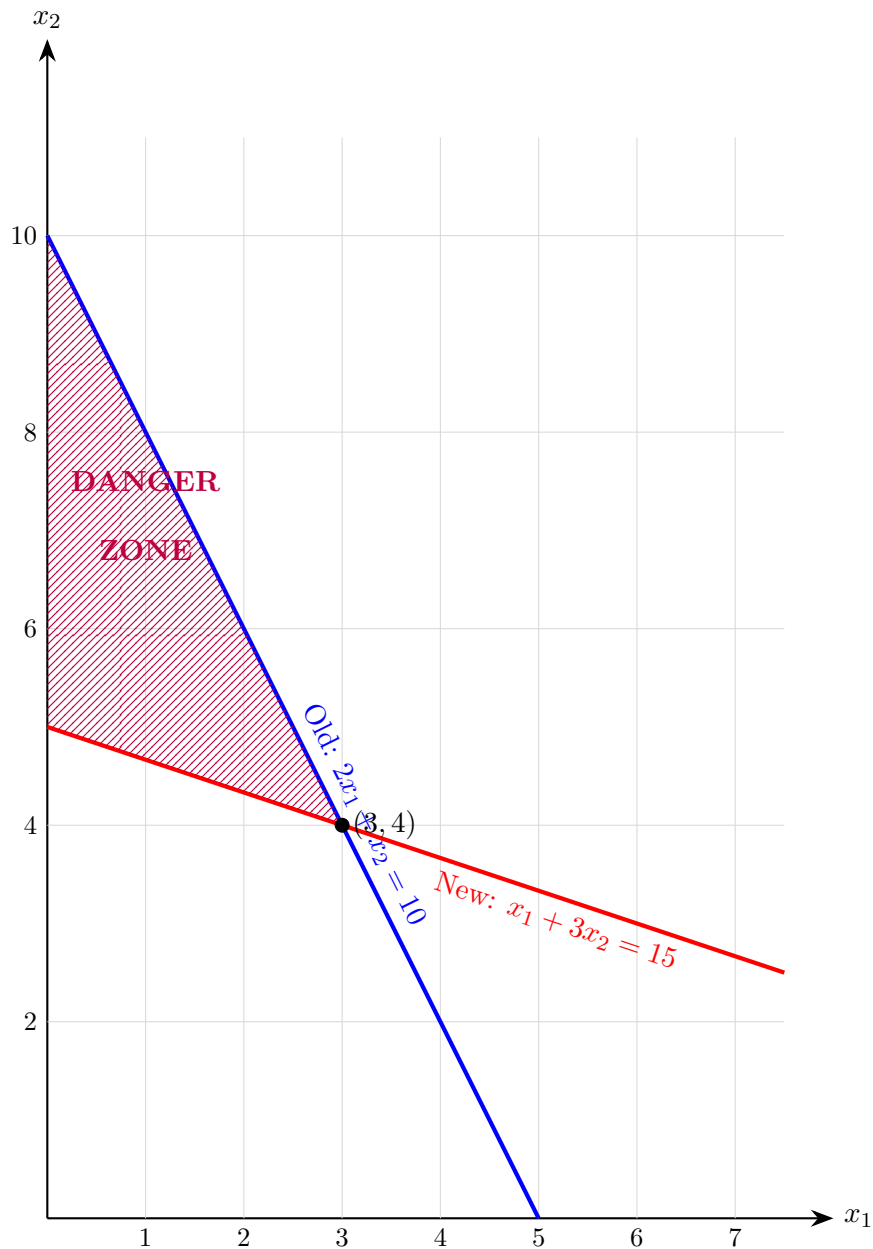
Characteristics:

- Triangle shape
- One vertex at original bundle $(4, 2)$
- Two vertices on x_1 -axis
- This happens when: old budget line is **steeper** than new iso-cost line

3 Scenario 2: Triangle NOT Touching x-axis

Setup:

- Old prices: (2, 1), bought (3, 4), budget = 10
- New prices: (1, 3), old bundle costs 15



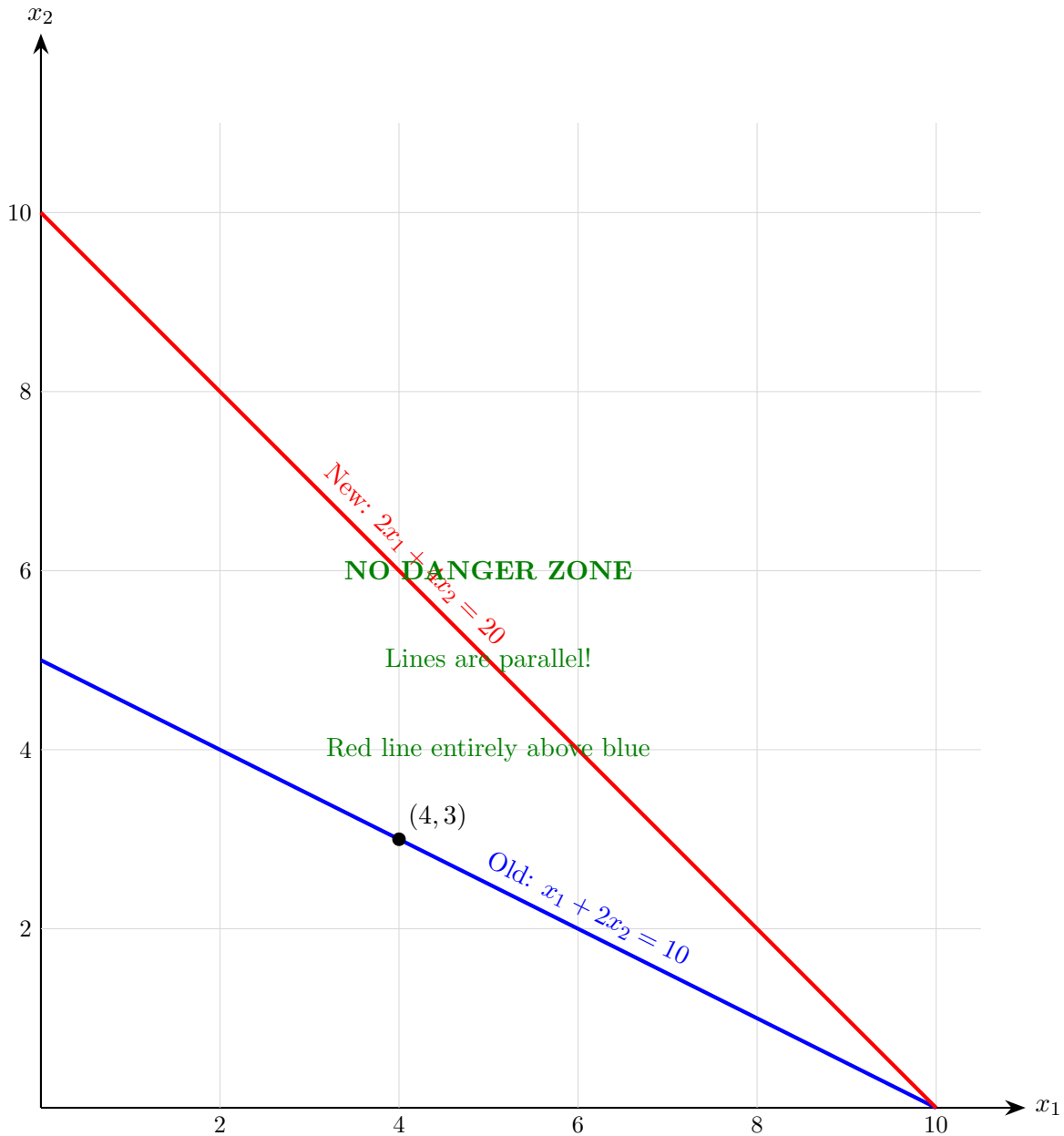
Characteristics:

- Triangle shape
- One vertex at original bundle (3, 4)
- Two vertices on x_2 -axis (NOT x_1 -axis!)
- This happens when: old budget line is **flatter** than new iso-cost line
- The danger zone is on the LEFT side instead of bottom

4 Scenario 3: NO Danger Zone (All Safe)

Setup:

- Old prices: (1, 2), bought (4, 3), budget = 10
- New prices: (2, 4), old bundle costs 20
- All prices **DOUBLED**!



Characteristics:

- The lines are **parallel** (same slope)
- New iso-cost line is entirely **above** old budget line
- No intersection = No danger zone

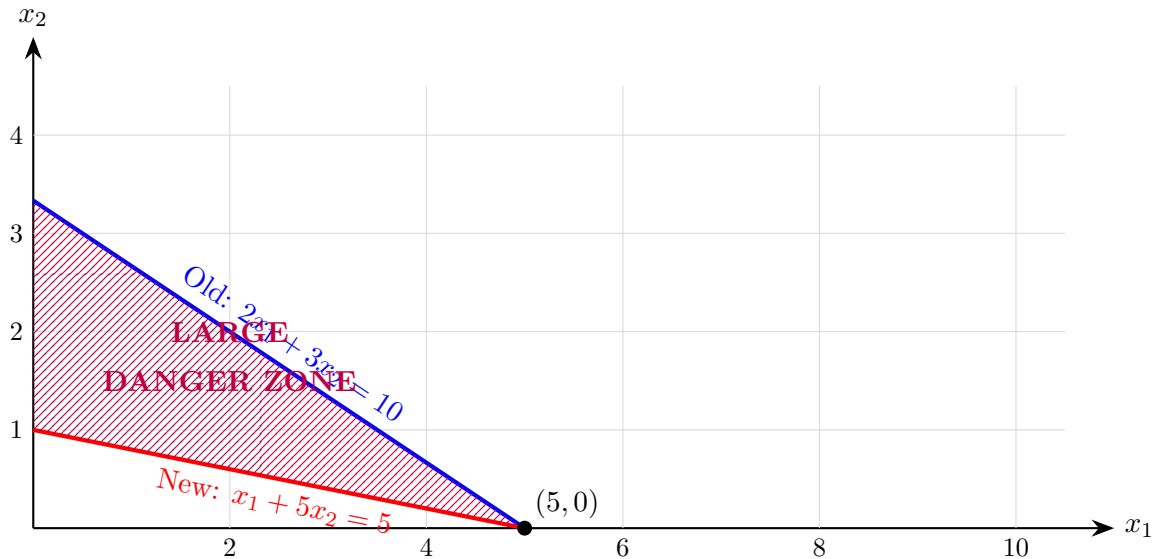
- **Every bundle is safe!**
- This happens when all prices change by the same proportion (multiply by constant)

Economic interpretation: If all prices double, any bundle you buy reveals your current budget is $2\times$ whatever it was. But that doesn't create a WARP violation because the affordability sets don't overlap properly.

5 Scenario 4: Entire Region is Danger Zone

Setup:

- Old prices: (2, 3), bought (5, 0), budget = 10
- New prices: (1, 5), old bundle costs 5
- Price of good 1 **decreased**, price of good 2 **increased**



Characteristics:

- Large triangular danger zone
- Original bundle is on the boundary (corner solution at (5, 0))
- New iso-cost line is **below** old budget line
- Most of what was affordable before creates violations
- This happens when: good 1 gets cheaper, good 2 gets more expensive, and you were buying only good 1

Why this is large: Since the new bundle (5,0) costs only \$5 at new prices, buying almost anything costs $\geq$ \$5, so most old affordable bundles create violations!

6 Summary Table

Scenario	Relative Slopes	Danger Zone Shape	Touches
Classic Triangle	Old steeper than new	Triangle below	x_1 -axis
Flipped Triangle	Old flatter than new	Triangle on left	x_2 -axis
No Danger Zone	Parallel lines (proportional prices)	None / Empty	Nothing
Large Zone	New line below old	Large triangle	Both axes

7 Key Insights

1. The danger zone is **not always a triangle touching the x-axis!**
2. The shape depends on:
 - Which price increased more (relative to the other)
 - The original bundle's composition
 - Whether prices changed proportionally
3. The danger zone can:
 - Touch x_1 -axis (when old budget steeper)
 - Touch x_2 -axis (when new iso-cost steeper)
 - Not exist at all (parallel lines)
 - Be very large (when new line below old)
4. **General rule:** The danger zone exists when the two lines intersect in the positive quadrant, and it's the region satisfying BOTH constraints.